

Characteristics of Odonata (Insecta) fauna of Köyceğiz-Dalyan Special Environmental Protected Area (SEPA) and its conservation

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ABSTRACT

1. Odonata fauna of Köyceğiz-Dalyan Special Environmental Protected Area (SEPA) comprises 28 species.
2. *Calopteryx splendens intermedia*, *Caliaeschna microstigma*, *Gomphus flavipes lineatus*, *Anax immaculifrons* and *Lindenia tetraphylla* belonged to threatened category while the other 23 species belonged to the unthreatened category (Least Concern) according to the criteria of the International Union for Conservation of Nature.
3. Odonata fauna of Köyceğiz-Dalyan Special Environmental Protected Area (SEPA) indicated the confluence of Palaearctic, Afrotropical and Oriental species.
4. The reasons of species richness of Odonata fauna are geological, zoogeographical, ecological and climatological properties of the region.
5. Turkey is one of the areas which comprising high concentration of threatened species and the highest number of endemics in Mediterranean areas.
6. The declaration placed Köyceğiz Dalyan region into Special Environmental Protected Area (SEPA) category, provided its protection against environmental destructions as a goal. Despite its conservation status, environmental destructions of the protected area could not be prevented.
7. Odonata fauna may use in monitoring programs of conservation status of Köyceğiz-Dalyan SEPA and its biodiversity.

KEY WORDS: Biodiversity, conservation, endangered species, Special Environmental Protected Area (SEPA), habitat quality, Köyceğiz-Dalyan, Mediterranean region, meromictic lake, Turkey, zoogeography.

Köyceğiz-Dalyan Özel Çevre Koruma Bölgesi (ÖÇKB)'nin Odonata (Insecta) faunasının özellikleri ve korunması

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ÖZ

1. Köyceğiz-Dalyan Özel Çevre Koruma Bölgesi (ÖÇKB)'nin Odonata faunası 28 türü kapsamaktadır.
2. Uluslararası Doğa Koruma Birliği'nin (International Union for Conservation of Nature, IUCN) kriterlerine göre *Calopteryx splendens intermedia*, *Caliaeschna microstigma*, *Gomphus flavipes lineatus*, *Anax immaculifrons* ve *Lindenia tetraphylla* tehlike altındaki türler kategorisine aitken, diğer 23 tür tehlike altında olmayan türler kategorisinde yer alır.
3. Köyceğiz-Dalyan Özel Çevre Koruma Bölgesi (ÖÇKB)'nin Odonata faunası Palaearctic, Afrotropical ve Oriental biyocoğrafik bölge türlerinin kesişme noktasındadır.
4. Odonata faunasının tür zenginliğinin nedeni bölgenin jeolojik, zoocoğrafik, ekolojik ve iklimsel özelliklerinin çeşitliliğidir.
5. Türkiye, Akdeniz bölgesinde çok sayıda tehlike altındaki türleri içeren ve yüksek endemizmin görüldüğü ülkelerden biridir.
6. Köyceğiz-Dalyan bölgesinin, Özel Çevre Koruma Bölgesi kategorisine dahil edilmesi, bölgenin çevresel tahribatlara karşı korunmasını bir hedef haline getirmiştir. Fakat bu koruma durumuna rağmen, bölgenin çevresel tahribatına engel olunamamıştır.
7. Odonata faunası, Özel Çevre Koruma Bölgesi'nin korunmasında ve bölgenin biyoçeşitliliğindeki değişimin izlenmesinde kullanılabilir.

ANAHTAR KELİMELELER: Akdeniz bölgesi, biyolojik çeşitlilik, habitat kalitesi, koruma, Köyceğiz-Dalyan, meromiktik göl, nesli tükenmekte olan türler, Özel Çevre Koruma Bölgesi (ÖÇKB), Türkiye, zoocoğrafya.

INTRODUCTION

Köyceğiz-Dalyan Special Environmental Protected Area (SEPA) shows a high diversity of plant and animal species. Aquatic ecosystem of Köyceğiz-Dalyan Special Environmental Protected Area (SEPA) is a complex of a wide variety of aquatic habitats such as running waters, brackish water and fresh water lakes, thermal and cold water springs, ground water sources, wetlands, floodplain forests, lagoon lakes, coastal lake, estuarine channel system and Mediterranean Sea. The habitat, species, genetic and metabolic diversity are very high in the area (Kazancı 1993).

Köyceğiz-Dalyan Special Environmental Protected Area (SEPA) is located in a region of considerable interest for heavy touristic activities, a process associated with excessive urbanization and loss of natural habitats.

For several years, Köyceğiz-Dalyan Special Environmental Protected Area (SEPA) has been the subject of considerable scientific research, conservation and protection because of its importance to biodiversity. Limnology and biodiversity of aquatic ecosystem of the area were detailly studied under Nilgün Kazancı's leadership for the first time between 1990 and 1993. This research project was supported by Deutsche *Gesellschaft für Technische Zusammenarbeit* (GTZ) GmbH and Water and Environmental Ministry of Turkey. The results of the projects were reported in various publications (Kazancı *et al.* 1992a, b, Kazancı 1993, Kazancı 1995, Kazancı *et al.* 1999a, b, Kazancı *et al.* 2000, Kazancı and Dügel 2000, Kazancı and Girgin 2001, Kazancı *et al.* 2008, Kazancı and Dügel 2010).

The first list of odonate fauna of the area was given by Kazancı (1995) with 28 species including new record of *Lindenia tetraphylla* from Turkey.

The knowledge of habitat properties of Odonata species is necessary step for their conservation activities in Turkey (Kazancı 2011). The particular habitat structure of Köyceğiz-Dalyan Special Environmental Protected Area is the main reason of species richness.

The present study was carried out to determine composition and characteristics of Odonata fauna of the Köyceğiz-Dalyan Special Environmental Protected Area (SEPA). Also some characteristics of protected area were reported. These knowledge can then help managers in the development of conservation and habitat quality monitoring plans for the area.

PROPERTIES OF STUDY AREA

Geographical location of Köyceğiz-Dalyan Special Environmental Protected Area

Köyceğiz-Dalyan SEPA (Figure 1) is located in Muğla province in southwestern part of Mediterranean Region, between 36°45' N and 37°15' N latitude and 28° 22' 30" and 28° 52' 30" E longitude. The whole protection area is 385km². The wide variety of aquatic habitats (four lagoon lakes, sulphiric thermal and karstic springs around and bottom of the Köyceğiz Lake, İztuzu coastal lake, the labyrinth-like Dalyan channel system between Köyceğiz Lake and Mediterranean Sea, six streams inflowing to the Köyceğiz Lake, floodplain forests of *Liquidambar orientalis*, wetlands and Mediterranean Sea) exist in the area. Centre of this complex aquatic ecosystem is the meromictic Köyceğiz Lake.

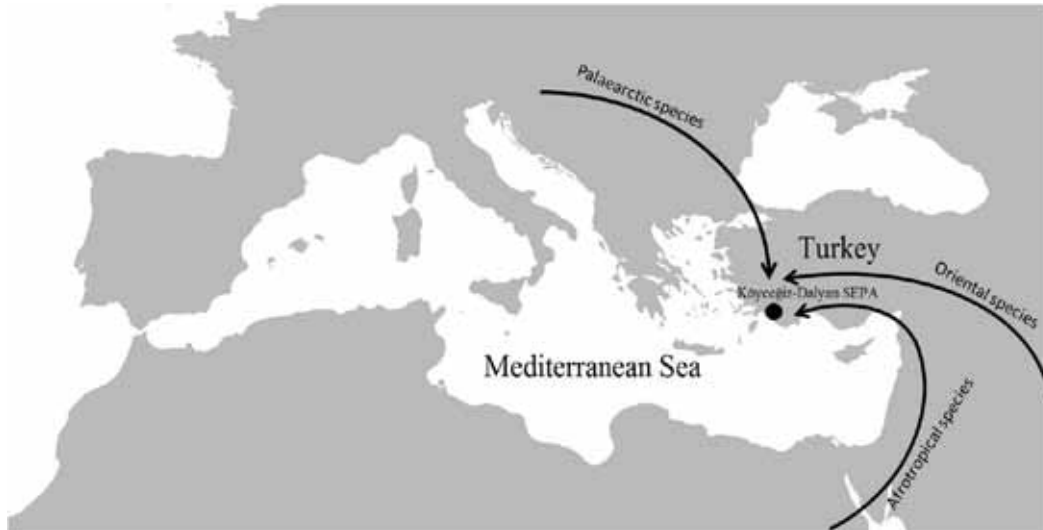


Figure 1. Köyceğiz-Dalyan Special Environmental Protected Area (SEPA).

Drainage area of Köyceğiz Lake and Dalyan channel system is 2100km². The area bordered to the north by mountains (2000m), to the south by Mediterranean Sea, to the west and east by hills (500-900m).

Geology

The geological structure has been given by Graciansky (1968). The mountain in the area belongs to Taurus mountain ridge of Alpine-Himalayan orogenic belt which was formed during the closure of Tethys Ocean. Köyceğiz Lake basin and river beds were formed during Plio-quaternary time. The allochthonous limestone is source of thermal, sulphiric and karstic springs. Other important water sources are Plio-quaternary alluvium which is bearing water originated from rainfall and from seepage of karstic limestone (Graciansky 1968).

Hydrology

Many springs with different water chemistry control the hydrology of the Köyceğiz-Dalyan aquatic ecosystem (Kazancı *et al.* 2008). Köyceğiz Lake is composed of two basins which are Köyceğiz and Sultaniye. The thermal groundwater sources are located in southern part of the lake basin (in Sultaniye basin) while karstic groundwater sources are located in northern part of the lake basin (in Köyceğiz basin). The lake is fed by six streams and by groundwater sources. Some of the running waters are intermittent streams and dry in summer period (Kazancı 1993, Kazancı and Dügel 2000). Maximum inflow of these streams to the lake is during winter and spring periods. Namnam Stream (drainage area is 253km², discharge is between 2.4 and 11m³/sec) and Yuvarlakçay Stream (discharge is 4 m³/sec) are perennial type and flow more than 90% of the time. These two streams are important water sources for mixolimnion of meromictic Köyceğiz Lake but the main water sources of the lake are sulphuric, thermal (30-40°C) groundwater sources and karstic cold groundwater sources in lake bottom of two basins (Kazancı 2003, Kazancı *et al.* 2008).

Climate

Climate of the area is Mediterranean type. While summer period is hot and dry, winter period is mild and rainy. The mean annual precipitation in the basin is 1202 mm. The mean annual temperature is about 18°C Kazancı *et al.* (1999 b, 2000).

Vegetation

Seçmen and Leblebici (1984), Güner *et al.* (1996) and Kazancı *et al.* (1999 b, 2000) studied on the flora of aquatic habitats in Köyceğiz-Dalyan SEPA. The vascular plant flora of the region contains 110 families, 470 genera and 924 species and belong to the following phytogeographic elements: i.e. Mediterranean (29.15%), Euro-Siberian (5.62%) and Irano-Turanian (1.79%).

Phragmites australis prevails lake littoral areas, Namnam Stream mouth and towards Dalyan channel. *Juncus maritimus* covers north and northwestern shore of Köyceğiz Lake and surrounding of Hamitköy channel. The development of aquatic vegetation of Köyceğiz-Dalyan SEPA depends on the salinity and water quality properties of aquatic habitats.

MATERIALS AND METHODS

Adults of Odonata species were caught with a hand net from all types of aquatic habitats. Nikon SMZ 10 trinocular was used for identifications of Odonata species. Conservation status of species are given according to IUCN Red List Categories and Criteria (2001).

RESULTS

Twenty eight species of Odonata were recorded (Table 1) from Köyceğiz-Dalyan SEPA. The number of Odonata species given from the area represents 1/5 of dragonflies and damselflies known from Turkey. The twenty eight species belong to eight families of Zygoptera and Anisoptera sub orders (Table 1). Eight species from Zygoptera and twenty species from Anisoptera were identified (Figure 2). Dominant families were Libellulidae (42,86%) and Aeshnidae (14,28%).

Calopteryx splendens intermedia, *Caliaeschna microstigma*, *Gomphus flavipes lineatus*, *Anax immaculifrons* and *Lindenia tetraphylla* belonged to threatened category while the other 23 species belonged to the unthreatened category (Least Concern) according to the criteria of the International Union for Conservation of Nature (IUCN 2001).

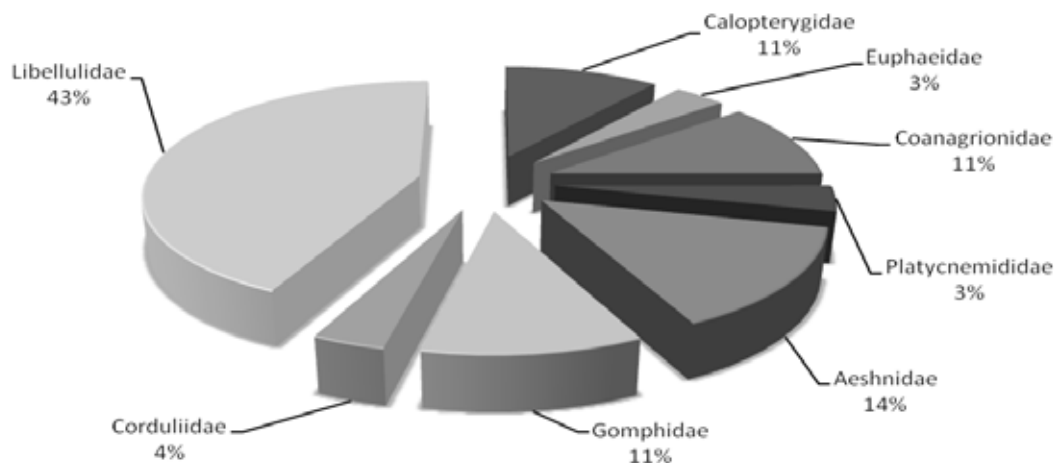


Figure 2. Percentages of Odonata by family in Köyceğiz-Dalyan SEPA.

DISCUSSION

Biogeography of Odonata collected in Köyceğiz-Dalyan SEPA

All recorded species belong to Palearctic biogeographic region (Table 1). Four species of them occur in Palearctic and Afrotropical biogeographic regions, four species of them occur in Palearctic, Afrotropical and Oriental biogeographic regions, one species of them occur in Palearctic and Oriental biogeographic regions. This result indicated the confluence of Palearctic, Afrotropical and Oriental species in Köyceğiz-Dalyan SEPA. Fauna of Odonata of Mediterranean region of Turkey comprises elements of Palearctic, Afrotropical, Oriental Zoogeographic Regions due to transitional position of Turkey between these zoogeographic regions (Kosswig 1955).

The zoogeographic composition of the Odonata fauna put emphasis on the importance of Köyceğiz-Dalyan SEPA for biodiversity. The reasons of high diversity of Odonata fauna are geological (mountains provide refuge areas and habitat diversity), zoogeographical (glacial periods), ecological (high variety of aquatic habitats) and climatological properties (Mediterranean climate) of the region. Taurus mountains settled from eastern to western part of southern Turkey. The position of mountain range provides distribution of Oriental and of Afrotropical species in the Mediterranean region. High diversity of the region is also related to the refuges created by Taurus mountain ranges. These refuges are harbour for isolated population pools of species in glacial periods (Kosswig 1955) and nowadays (Kazancı 1991). Stable conditions of Mediterranean climate and sub tropical properties of Mediterranean climate support the high species diversity of odonate species. High variety of physical and chemical properties of aquatic and terrestrial habitats in the Köyceğiz-Dalyan SEPA are other reasons of high species richness of odonate fauna.

The Family Libellulidae (42,86%) is well represented in the Köyceğiz-Dalyan SEPA (Figure 2). Libellulidae species have high distribution capacity and one of the largest odonate families in the world (Kalkman *et al.* 2008).

Protection of the Köyceğiz-Dalyan SEPA and its Odonata Fauna

The insect conservation is closely related with conservation of biodiversity. Establishment of protected areas is seen to be the most important strategy for biodiversity conservation (Pimm *et al.* 2001). Despite establishment of protected areas many ecosystems remain poorly protected (Hoekstra *et al.* 2005).

Special Environmental Protected Area (SEPA) is a combination of biologically sensitive areas within populated landscapes and human settlements. SEPAs were defined as “areas of land, with coast and sea as appropriate, where the interaction of people and nature over time has produced an area of distinct character with significant aesthetic, ecological and/or cultural value, and often with high biological diversity”, and classified as “Category V, Managed Resource Protected Area Protected Landscape/Seascape” by The International Union for Conservation of Nature and Natural Resources (IUCN).

Köyceğiz-Dalyan region has all these properties with Kaunos ruins dated BC 18 and with its high biodiversity. Köyceğiz-Dalyan region was declared as a Special Environmental Protected Area ‘SEPA’ by the “Environment Protection Agency for Special Areas (EPASA)” on 1988. Its boundaries were extended in 1990.

The declaration placed Köyceğiz Dalyan region into SEPA category, provided its protection against environmental destructions as a goal. Despite its conservation status, to attempt reconstruction of hydroelectric power plants on the running waters could not be prevented.

Odonata fauna may be used in monitoring programs of conservation status of Köyceğiz-Dalyan SEPA and its biodiversity. Aquatic insects are very sensitive to habitat degradation and remaining alive in relation to the richness of undisturbed aquatic habitats.

According to Odonata fauna of Köyceğiz-Dalyan SEPA it can be regarded that it has very rich fauna and a refuge area for threatened species. The main reason of this richness is various types of aquatic habitats in the area by means of different chemical, physical and biological characteristics.

Odonata fauna of Köyceğiz-Dalyan SEPA contains almost 1/5 of Turkish Odonata fauna with 28 species. Five species in the area, *Calopteryx splendens intermedia*, *Caliaeschna microstigma*, *Gomphus flavipes lineatus*, *Anax immaculifrons* and *Lindenia tetraphylla* are classified as Near Threatened (NT) in Red List and the population trends of these species are stated as decreasing in Mediterranean region (IUCN 2001, Riservato, *et al.* 2009). Other 23 species are listed in LC (Least Concern) category.

Myers *et al.* (2000) identified 25 hotspots in the world according to concentration of endemism, species richness and degree of threat in an area and Mediterranean Region is one of these areas.

Riservato *et al.* (2009) reported conservation status of 165 Mediterranean species of Odonata from the Mediterranean basin with the IUCN regional Red List criteria (IUCN 2001). According to the results of this study, Turkey is one of the areas which comprising high concentration of threatened species and the highest number of endemics. Turkey has the high biodiversity of Odonata with other species rich Mediterranean areas (Italy, France, the Balkans region, Greece and Tunisia).

Pollution, climate change, severe weather conditions, residential, touristic and commercial development, agriculture and aquaculture, energy production (Hydroelectric power plant construction on the streams), transportation (motor boats in Dalyan channel system), human intrusions and disturbance and natural system modifications are significant threats to odonate fauna in the Köyceğiz-Dalyan SEPA.

Tablo 1. Species List, IUCN Category, endemism, distribution areas and zoogeographic regions.

Order	Family	Percentages	Species	IUCN Category	Endemism	Distribution areas and biogeographic regions
Zygoptera	Calopterygidae	10,71%	<i>Calopteryx splendens intermedia</i> Sel.	NT	Not M.E.*	European-Mediterranean group Palearctic-Mediterranean
			<i>Calopteryx splendens</i> ssp. (=hybrid " <i>splendens</i> " X <i>intermedia</i>)	LC	"	European-Mediterranean group Palearctic-Mediterranean
			<i>Calopteryx virgo festiva</i> (Brullé)	LC	"	Balkan-Mediterranean (Turkey), Palearctic
	Euphaeidae	3,57%	<i>Epallage fatima</i> Charp.	LC	"	Balkan, Mediterranean, Middle east, Palearctic
	Coanagrionidae	10,71%	<i>Erythromma l. lindenii</i> (Sel.)	LC	"	Europe, Mediterranean, Middle east, Palearctic
			<i>Coenagrion puella</i> (L.)	LC	"	Europe, Mediterranean, Middle east, Palearctic
			<i>Ischnura elegans ebneri</i> Schmidt	LC	"	Mediterranean, Middle east, Palearctic
	Platycnemididae	3,57%	<i>Platycnemis p. pennipes</i> (Pall.)	LC	"	European-Mediterranean, Middle east, Palearctic
Anisoptera	Aeshnidae	14,28%	<i>Caliaeschna microstigma</i> (Schneider)	NT	"	European-Mediterranean group Palearctic-Mediterranean Middle east
			<i>Anaciaeschna isosceles antehumeralis</i> (Schneider)	LC	"	European-Mediterranean group Palearctic-Mediterranean Caucasus Greece Turkey
			<i>Anax immaculifrons</i> Ramb.	NT	"	Mediterranean, Middle east, Palearctic-Oriental
			<i>Anax imperator</i> Leach	LC	"	Palearctic-Afrotropical-Oriental
	Gomphidae	10,71%	<i>Gomphus flavipes lineatus</i> Bart.	NT	"	Mediterranean, Middle east, Palearctic
			<i>Gomphus schneideri</i> Sel.	LC	"	Balkan, Mediterranean, Middle east, Palearctic
			<i>Lindenia tetraphylla</i> (Vander L.)	NT	"	Mediterranean, Middle east, Palearctic, Afrotropical
	Corduliidae	3,57%	<i>Somatochlora flavomaculata</i> (Vander L.)	LC	"	Balkan, Mediterranean, Middle east, Palearctic
	Libellulidae	42,86%	<i>Libellula fulva</i> L.	LC	"	European, Mediterranean, Middle east, Palearctic,
			<i>Orthetrum b. brunneum</i> (B. de Fonsc.)	LC	"	European-Mediterranean group Palearctic-Mediterranean
			<i>Orthetrum c. chrysostigma</i> (Burm.)	LC	"	Mediterranean, Palearctic, Afrotropical
			<i>Orthetrum ramburi</i> (Sel.)	LC	"	European-Mediterranean group Palearctic-Mediterranean
			<i>Orthetrum sabina</i> (Dru.)	LC	"	Europe, Mediterranean, Middle east, Palearctic, Afrotropical, Oriental
			<i>Orthetrum taeniolatum</i> (Schneider)	LC	"	Europe, Mediterranean, Middle east, Palearctic, Afrotropical, Oriental
			<i>Crocothemis erythraea</i> (Brullé)	LC	"	Europe, Mediterranean, Middle east, Palearctic, Afrotropical, Oriental
			<i>Sympetrum f. flaveolum</i> (L.)	LC	"	European-Mediterranean group Palearctic-Mediterranean
			<i>Sympetrum fonscolombii</i> (Sel.)	LC	"	Europe, Mediterranean, Middle east, Palearctic, Afrotropical
			<i>Sympetrum meridionale</i> (Sel.)	LC	"	Europe, Mediterranean, Middle east, Palearctic, Afrotropical
			<i>Sympetrum striolatum</i> (Charp.)	LC	"	European-Mediterranean group Palearctic-Mediterranean
			<i>Trithemis festiva</i> (Ramb.)	LC	"	Mediterranean, Middle east, Palearctic

* Not M.E.: Not Mediterranean Endemic, NT: Near Threatened; LC: Least Concern

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